

Mehrdad Momeni Zadeh

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TECHNICAL SKILLS

Languages: Python, Java, C++, Go, TypeScript, SQL, Kotlin, Ruby
Web & Backend: Spring Boot, Hibernate, NestJS, Next.js, React, Django, RESTful APIs, GraphQL
ML & Data: PyTorch, TensorFlow, scikit-learn, OpenCV, WandB, Hydra, NumPy, Pandas
DevOps & Cloud: Docker, Kubernetes, Jenkins, Git, CI/CD, Kafka, GCP, AWS (S3, EC2), JWT/OAuth2
Databases: PostgreSQL, MySQL, MongoDB, Firebase, H2

WORK EXPERIENCE

DeLTA Lab — Simon Fraser University Burnaby, BC
Undergraduate Researcher — Scientific Machine Learning Dec 2024 – May 2025

- **Second author** on **NeurIPS 2025** paper, resubmitted to **ICLR 2026** on data-efficient neural operators.
- Boosted PDE benchmark accuracy by **12%** and cut training cost **4×** on Google Cloud TPU-v4.
- Built forecasting workflows that lowered long-horizon error **15–20%**.
- Packaged experiments in **Docker**, reducing setup time from **1 hr** to **10 min**.
- Integrated Git + CI/CD pipelines, cutting integration bugs by **30%** and speeding up team experiment cycles by **2×**.

TECHNICAL PROJECTS

StagePass — Full-stack event ticketing system for university events. GitHub 🔗
Tech: Java, Spring Boot 3, Spring Security, TypeScript, OAuth2, PostgreSQL, React, Stripe, REST APIs

- Deployed at **SFU's 2025 Events**, processing **100+** student tickets with zero payment errors.
- Secured organizer/attendee flows via **Spring Security** + OAuth2, ensuring role-based access.
- Designed **REST APIs** and **PostgreSQL** schema to handle **300+ tickets/day**.

ShareScreenly — Platform for screen recording and instant video sharing. GitHub 🔗
Tech: TypeScript, Next.js 15, Node.js, PostgreSQL, Tailwind, AWS S3, FFmpeg

- Adopted by **100+** SFU students, cutting demo setup from **5 min** to **30 sec**.
- Integrated **AWS S3** + **FFmpeg** for storage, compression, and metadata; supported **1GB+** uploads.
- Shipped role-based access with **Better Auth** and a responsive **Tailwind** UI.

MicroGopher — Distributed microservices platform in Go. GitHub 🔗
Tech: Go, Docker, Kubernetes, PostgreSQL, MongoDB, gRPC, AMQP

- Split a monolith into modular **Go microservices**, improving fault isolation + scalability.
- Built services for auth, logging, and mailer communicating via **REST APIs** + **gRPC**.
- Deployed services with **Docker** + **Kubernetes**, handling **10K+ req/sec** with under **1%** errors.

HirePrep — AI-powered mock interview platform with real-time voice practice. GitHub 🔗
Tech: TypeScript, Next.js 15, Tailwind, Firebase, Gemini AI, Vapi, ShadCN

- Piloted with **50+ SFU students**, improving interview scores by **20%** in feedback surveys.
- Built secure authentication + session management with **Firebase** + **Firestore**.
- Generated tailored interview questions and instant **AI-driven feedback** using Gemini.

NeuroDriver — 2D driving simulator with neural agents. GitHub 🔗
Tech: Python, Box2D, Genetic Algorithms, NumPy

- Trained neural agents to reduce collisions by **20%** over **50 generations**.
- Improved lap completion rates by **30%** through reward-tuned training.
- Automated **hyper-parameter sweeps** + GIF replays via Makefile pipeline.

EDUCATION

BSc. in Computer Science — GPA: 3.4/4.33 Expected Dec 2026
Simon Fraser University Burnaby, BC

- **Relevant Coursework:** Data Structures & Algorithms, Computer Systems, Databases, Networking, Machine Learning, Deep Learning, Computer Vision, Human-Computer Interaction